

DATA_1200-02 - INTRODUCTION TO DATA ANALYSIS

ASSIGNMENT: 1

Linear Regression in Python



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1 Objective: -

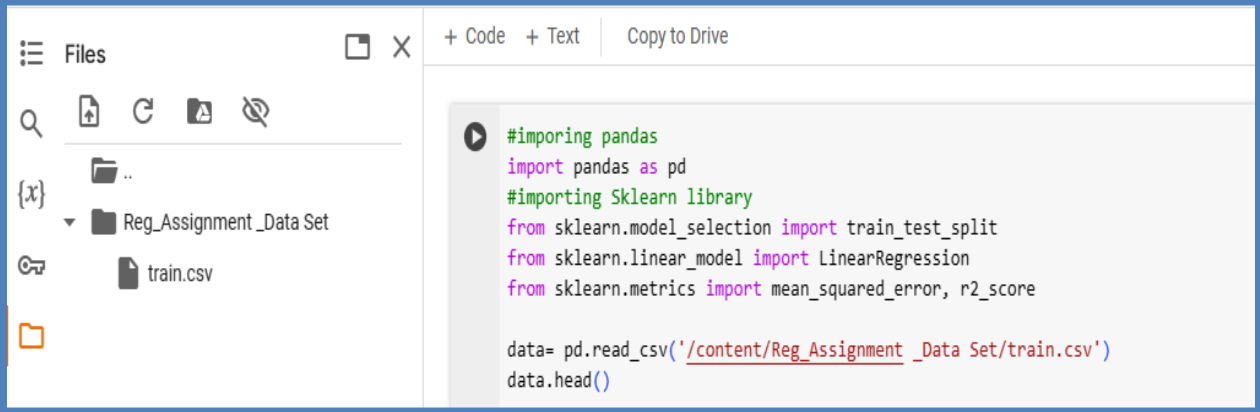
In the following, I am trying to predict housing prices based on the "OverallQual" feature using linear regression which is a statistical method used to model the relationship between a dependent variable (housing prices) and one or more independent variables (features-OverallQual).

2 Dataset Acquisition: -

I have downloaded the dataset from [Kaggle]([House Prices - Advanced Regression Techniques | Kaggle](#)) and loaded it into a pandas dataframe.

3 Exploratory Data Analysis (EDA):-

Performed below EDA to understand the dataset's features.



The screenshot shows a Jupyter Notebook interface. On the left, the 'Files' pane displays a directory structure with a folder named 'Reg_Assignment_Data Set' containing a file 'train.csv'. The main area shows two code cells. The first cell contains imports for pandas, sklearn's train_test_split, LinearRegression, mean_squared_error, and r2_score. The second cell contains the code to read the CSV file and display its first few rows.

```
#importing pandas
import pandas as pd
#importing Sklearn library
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

data= pd.read_csv('/content/Reg_Assignment _Data Set/train.csv')
data.head()
```

- **Data Cleaning:-**
 - Handled the missing values in the columns.

```
# Dropped the columns with more than 40% missing values
data_cleaned = data.dropna(thresh=len(data)*0.4,axis=1)
data.head()
```

- **Split the Dataset:-**
 - Separate the columns(OverallQual & SalePrice) into features (X) and label or Target (y) and split the dataset into training and testing sets.

```
#Separated as Feature and Label
X = data_cleaned[['OverallQual']]
y = data_cleaned[['SalePrice']]
# Split the dataset into features (X) and labels (y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=80)
```

- **Implement Linear Regression Model:-**

- Imported and trained the linear regression model

```
# Train the linear regression model
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

- **Model Evaluation**

- Evaluated the model on the test set using Mean Squared Error (MSE) and R-squared.

```
print(f'Coefficient of determination is {r2_score(y_test, y_pred)}')
```

- ✚ **Result :** The coefficient of determination is 0.70114206738856. Which represents linear regression between two variables (SalePrice and OverallQuality)

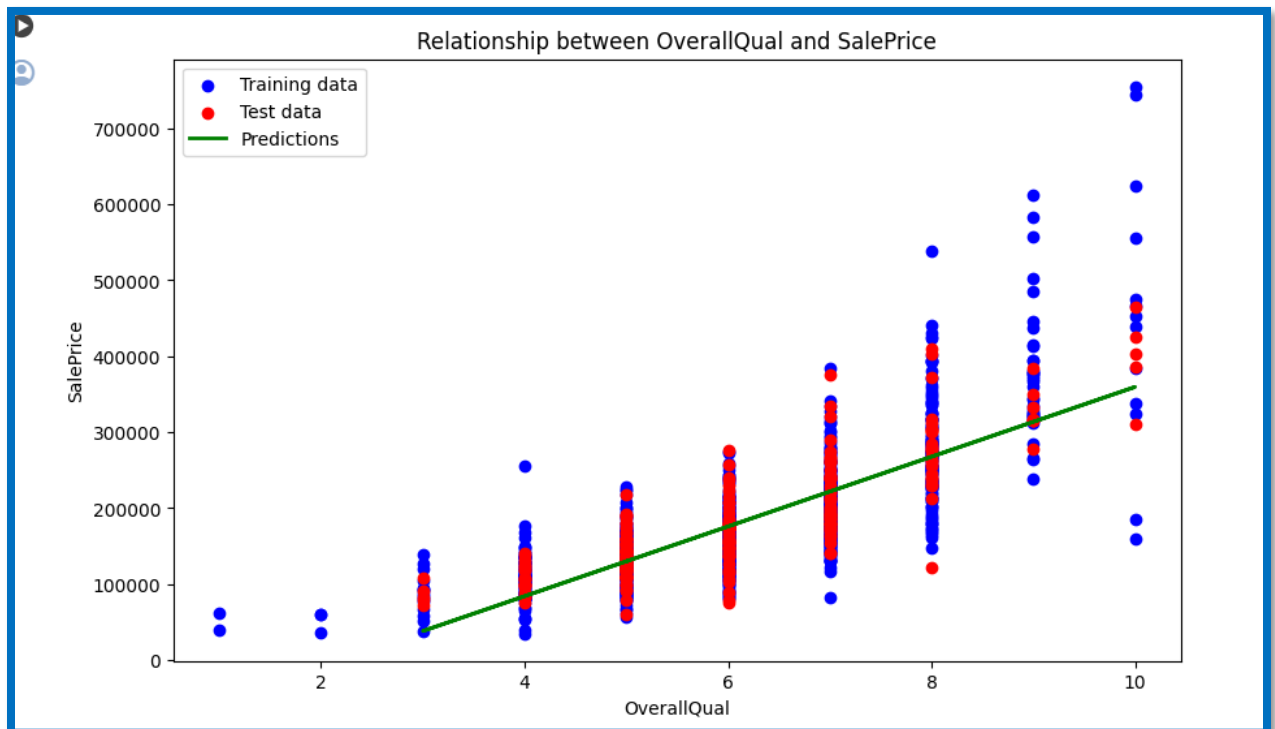
```
Coefficient of determination is 0.7011420673885607
```

- **Visualizing the Linear Regression Model using the scatter plot.**

- I have imported the matplotlib.pyplot library as plt and represented the Linear Regression Model.

```
import matplotlib.pyplot as plt
# Plotting the scatter plot
plt.figure(figsize=(10, 6))
plt.scatter(X_train, y_train, color='blue', label='Training data')
plt.scatter(X_test, y_test, color='red', label='Test data')
plt.plot(X_test, y_pred, color='green', linewidth=2, label='Predictions')
plt.xlabel('OverallQual')
plt.ylabel('SalePrice')
plt.title('Relationship between OverallQual and SalePrice')
plt.legend()
plt.show()
```

Visualization :



- Predict the Housing Pricing using a linear regression model for the Overall Quality.

```
# Predict the housing price for a new OverallQual value

new_overall_qual = [[11]]
predicted_price = model.predict(new_overall_qual)
print(f'Predicted housing price for OverallQual 11 is {predicted_price[0][0]}')
```

Predicted housing price for OverallQual 11 is 405549.6457521691

🚩 **Result :** Predicted housing price for OverallQual(OverallQuality) 11 is 405549.6457521691

Over All Code :

```
#importing pandas
import pandas as pd
#importing Sklearn library
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

data= pd.read_csv('/content/Reg_Assignment _Data Set/train.csv')
data.head()

# Dropped the columns with more than 40% missing values
data_cleaned = data.dropna(thresh=len(data)*0.4,axis=1)
data.head()

#Seperated as Feature and Label
X = data_cleaned[['OverallQual']]
y = data_cleaned[['SalePrice']]
# Split the dataset into features (X) and labels (y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=80)

# Train the linear regression model
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```

print(f'Coefficient of determination is {r2_score(y_test, y_pred)}')

import matplotlib.pyplot as plt
# Plotting the scatter plot
plt.figure(figsize=(10, 6))
plt.scatter(X_train, y_train, color='blue', label='Training data')
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plt.xlabel('OverallQual')
plt.ylabel('SalePrice')
plt.title('Relationship between OverallQual and SalePrice')
plt.legend()
plt.show()

# Predict the housing price for a new OverallQual value

new_overall_qual = [[11]]
predicted_price = model.predict(new_overall_qual)
print(f'Predicted housing price for OverallQual 11 is
{predicted_price[0][0]}')

```

Google Collab Link to access the code:

https://colab.research.google.com/drive/19B8-2gw7PH3TDkPP0_XTeFr1uK6Ehhmm?usp=sharing

4 Problem tried to solve and Conclusion :-

- ✓ The problem addressed in this analysis was to predict housing Saleprices based on the overall quality of the property. The dataset used contained information about various features of residential homes, including the overall quality rating, which ranged from 1 to 10. The goal was to develop a model that could accurately predict housing prices based on this single feature.
- ✓ The analysis utilized a linear regression model to predict housing Saleprices based on the overall quality rating. The coefficient of determination (R-squared) value was found to be approximately 0.701, indicating that the model explains about 70.1% of the variability in housing prices using the overall quality rating alone. This suggests that overall quality is a significant predictor of housing prices, but there are likely other factors that also influence prices.

- ✓ Using the developed model, the predicted housing price for a property with an overall quality rating of 11 was approximately \$405,549.65. While this prediction provides a rough estimate, it's important to note that there may be some variability or error in the prediction.
- ✓ In conclusion, the analysis demonstrates that overall quality is an important factor in determining housing prices, and the developed model shows moderate success in predicting prices based on this feature alone. Further analysis could explore additional features to improve the model's accuracy and reliability.